

# **Segmented and Individualized Discount Optimization in Email Marketing**

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# Abstract

In today's highly competitive e-commerce environment, brands are facing increasing challenges in balancing short-term sales growth with long-term profitability. Over reliance on discounting risks eroding brand equity and wasting marketing budgets on customers who would convert without discounts.

This research analyzes how personalized discount allocation can be optimized to improve both profitability and customer targeting in email marketing. Using customer data from a company located in Europe that focuses on selling sportswear, the study develops a step-by-step approach, which first groups customers into meaningful segments and then evaluates the expected outcomes of offering different discount levels to each group. To avoid unnecessary discount spending, a set of rules is introduced to identify customers who are likely to respond positively to discounts and convert. The approach allows discounts to be allocated more efficiently, offering discounts only when they are expected to encourage additional purchases rather than targeting customers who would have purchased anyway. Simulation results suggest that this strategy can achieve higher profits while using significantly less discount budget, with only a small impact on overall sales volume. The findings also show differences in how customer groups respond to discounts, offering practical guidance for managers on how to balance short-term sales goals with long-term customer relationships and brand value.

The research offers practical guidelines for managers to implement more efficient discount targeting, highlighting the importance of balancing commercial gains with brand health. The findings demonstrate how businesses can move beyond generalized promotions toward more sustainable, personalized marketing strategies, even when randomized experimentation is infeasible.

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# 1 Introduction

## 1.1 Motivation

“The retailer has trained its customers to wait for discounts.” (Chad, 2018) This quote, widely shared in the retail industry, highlights a common problem in modern consumer marketing: the overreliance on discounting. Brands regularly run promotions through email marketing, which successfully trigger short-term conversions but also to some extent fuel a pattern of customer behavior that is harmful in the long run (Zeelenberg and Putten, 2005). As customers come to expect discounts, full-price purchases decrease, base sales decline, and brand equity erodes. This effect, sometimes referred to as the “Performance Ostrich Effect” (Andrew, 2025), traps brands in a cycle of performance spikes during promotions followed by stagnation.

At the same time, a growing scale of academic research and commercial tools in uplift modeling and causal inference offer a possible way out. Machine learning models now help marketers to estimate the incremental impact of each campaign or treatment on an individual level. This concept is formalized as the Conditional Average Treatment Effect (CATE) (Wager and Athey, 2018). While uplift modeling has attracted growing interest in fields such as personalized medicine and political campaigning, most developments in marketing have so far focused on direct marketing and response modeling, with opportunities remaining for broader applications in areas like customer relationship management and email targeting. (Devriendt et al., 2018) Companies like Nike have begun exploring personalized marketing through AI-driven personalization (Technosoft, 2025), but for widespread adoption in discount targeting, there is still a long way to go.

In an era where promotions are both a necessity and a liability, retailers face a pressing need to identify which customers truly require incentives and which do not. This thesis responds to that need by proposing a data-driven, causally informed framework to optimize discount targeting in email marketing campaigns.

This research is conducted in collaboration with a company located in Europe that focuses on selling sportswear. As the brand continues to grow its presence across both e-commerce and physical retail, optimizing customer engagement and promotional efficiency through channels like email marketing is increasingly important. The company’s scale, marketing maturity, and access to transactional and behavioral data provide an environment to test and evaluate personalized discount strategies in a real-world setting.

## 1.2 Research Questions

This research addresses the challenge of optimizing discount allocation in email marketing by balancing purchase stimulation and discount cost efficiency. While many promotional campaigns apply generalized discount offers, this study investigates how personalized

discount recommendations can improve revenue outcomes while minimizing unnecessary promotional cost.

To explore this goal, the following research questions are formulated:

**RQ1:** *How can personalized discount policies be developed under incomplete treatment exposure data using causal inference approaches?*

**RQ2:** *How can uplift modeling and counterfactual simulations inform optimal discount allocation policies at both segment and individual levels?*

**RQ3:** *What are the business impacts of implementing the optimized discount policy, in terms of revenue uplift, profit gain, purchase conversion, and discount cost reduction?*

### 1.3 Managerial Relevance

The findings of this research directly address a growing concern among CRM, pricing, and marketing teams: how to make promotional efforts more efficient and customer-centric. At the company, promotional emails with discount offers are typically sent to customers selected by purchase frequency in a recent period. While this strategy aims to capitalize on a high probability of conversion, it potentially results in over-targeting heavy buyers, those who might have made a purchase regardless of receiving a discount. In doing so, the brand potentially sacrifices profit margins without generating truly incremental value (Dubé and Misra, 2023).

However, a significantly larger group of light buyers, customers who may purchase less frequently but are not lost, remains under-addressed. These customers might not respond to one-size-fits-all discounting, but could be driven to convert with personalized and appropriately sized offers. Saving discount spend on high-propensity buyers opens the opportunity to reallocate resources toward reaching this broader audience via omnichannel strategies. By targeting more customers at a lower overall cost, the brand can potentially drive more incremental revenue and margin.

By introducing data-driven personalization, the company could allocate promotional budgets more effectively, preserve brand value by minimizing over-discounting, avoid training customers to wait for discounts, and improve overall marketing return on investment (ROI). Stakeholders across marketing operations, analytics, and executive leadership at the company stand to benefit from an approach that offers better control over promotional strategy, customer experience, and long-term profitability.

### 1.4 Academic Relevance

This thesis contributes to the intersection of marketing analytics, causal machine learning, and CRM strategy. It builds on literature related to uplift modeling, heterogeneous treatment effect estimation, and policy evaluation under observational settings (Wager and Athey, 2018; Austin and Stuart, 2015; Hirano et al., 2003). Unlike existing studies that

focus primarily on binary treatments or theoretical simulations, this research engages with real-world complexities such as multivalued treatments, inferred negatives, and seasonal fixed effects.

It also introduces a practical application of off-policy evaluation using IPW to simulate and assess targeting policies. By doing so, this study helps bridge the gap between methodological rationality and practical implementation in marketing decision-making.

## 1.5 Thesis Structure

The structure of this thesis reflects the overall flow of the research. The thesis begins by reviewing relevant literature on discounting strategies, personalized targeting, customer segmentation, treatment effects in marketing, and evaluation of targeting strategies without actual execution. Following this, the methodology chapter describes the data sources, preprocessing procedures, and the modeling framework adopted to estimate the causal effects of discounts and simulate alternative targeting policies. The proposed approach combines customer segmentation, propensity score estimation, causal inference modeling, and off-policy evaluation to assess the business impact of discount strategies.

The results chapter presents the empirical findings, including descriptive patterns in the data, customer heterogeneity, simulation outcomes of the optimized targeting policy, and sensitivity analyses to validate the robustness of the results. Finally, the thesis concludes with a discussion of the strategic implications for managerial decision-making, reflections on model performance and business trade-offs, and potential limitations and directions for future research.

# 2 Theoretical Background

## 2.1 The Effects of Over-Discounting

Discounting is widely used in marketing to stimulate immediate purchases, attract customers, and enhance short-term sales performance. However, frequent and significant discounting practices may carry substantial long-term drawbacks, undermining profitability and brand equity (Zeelenberg and Putten, 2005; Van Heerde et al., 2000). This phenomenon, commonly known as the "post-promotion dip", refers to a temporary decline in product sales immediately following a promotional period (Van Heerde et al., 2000). While conventional explanations attribute this decline primarily to consumer stockpiling and forward buying, Zeelenberg and Van Putten (Zeelenberg and Putten, 2005) suggest an important mechanism, brand switching, triggered by missed discounts.

Their empirical findings indicate that consumers show significant switching behavior when they miss deep discounts, potentially resulting in a permanent loss of brand loyalty if alternative brands satisfy their needs (Zeelenberg and Putten, 2005). This behavior

aligns with the psychological concept of "inaction inertia", where missing an attractive initial discount reduces the likelihood of consumers taking subsequent purchasing actions (Tykocinski and Pittman, 1998). Essentially, a large missed discount diminishes the attractiveness of future offers, encouraging consumers to explore alternative brands (Zeelenberg and Putten, 2005).

Moreover, heavy discounting can negatively affect consumer perceptions of brand value and product quality. Research in behavioral economics and consumer psychology suggests that repeated deep discounts can lead customers to develop lower reference prices, making them unwilling to purchase at regular prices in the future (Winer, 1986). Such price-driven consumer behavior can erode brand equity over time, resulting in decreased willingness to pay premium prices and weakened long-term profitability.

Additionally, extensive discounting often targets existing loyal customers who would likely purchase even without discounts, creating unnecessary promotional costs and undermining the company's profitability. Conversely, neglecting the role of state dependence in consumer choice behavior could prevent brands from optimizing pricing and targeting strategies (Dubé, Hitsch, et al., 2010). Therefore, a thoughtful and strategic approach to discount management is necessary to mitigate these negative effects.

Given these considerations, marketers must strategically manage promotional intensity and frequency, ensuring they balance short-term sales benefits against potential long-term brand damage and profitability erosion. This thesis aims to address these concerns by exploring optimized, personalized discounting strategies, aligning short-term promotional objectives with sustained long-term profitability and brand equity.

## 2.2 Personalized Coupon Targeting

Personalized coupon targeting leverages consumer-specific data to deliver tailored promotional offers, aiming to maximize customer engagement and profitability. This approach contrasts traditional mass discounting strategies by providing consumers with individualized incentives aligned with their purchase history, preferences, and predicted behavior (Zhang and Krishnamurthi, 2004). By aligning promotions closely with consumer needs and interests, personalized coupon targeting significantly enhances promotional efficiency, reducing unnecessary costs associated with broad, non-targeted discounts (Smith et al., 2023).

Dynamic coupon targeting strategies, which leverages real-time data and predictive analytics, are particularly promising. Liu (X. Liu, 2023) highlights the effectiveness of dynamic coupon targeting implemented through advanced machine learning and reinforcement learning methods. Such approaches dynamically adjust promotional offers based on continuously updated consumer behavior data, optimizing both short-term revenue and long-term customer value.



Personalized targeting can be enriched through predictive analytics and artificial intelligence, which anticipate customer behavior with greater accuracy (Ban and Keskin, 2021). AI-driven predictive models help firms to design highly personalized marketing strategies that not only target the right customers but also deliver promotions at optimal timing and scale, thereby substantially improving customer engagement and revenue performance (Chen et al., 2022; Elmachtoub et al., 2021).

The advantages of personalized coupon targeting extend beyond immediate revenue gains. Dubé and Misra (Dubé and Misra, 2023) highlight that personalized pricing can better align discounts with individual purchasing preferences, allowing many consumers to benefit comparing to uniform pricing, and targeted promotions offer consumers great benefits by better aligning price discounts with their actual purchasing preferences. This targeted approach thus contributes to a sustainable relationship between consumers and brands, developing loyalty without unnecessarily eroding brand equity or profitability.

The successful implementation of personalized coupon targeting requires robust analytical capabilities to leverage high-dimensional customer data (Ban and Keskin, 2021; Smith et al., 2023). Firms must also address operational challenges such as data quality, integration, and privacy to harness the potential benefits of personalized marketing strategies. This thesis investigates these dimensions by exploring and validating advanced personalized coupon targeting methodologies within the context of email marketing, specifically for the company.

## 2.3 Customer Segmentation

Customer segmentation is critical in maximizing the effectiveness of personalized marketing strategies. Effective segmentation methods can identify distinct customer groups based on their purchasing behaviors, preferences, and potential profitability, enabling businesses to tailor their marketing strategies accordingly (Tsai et al., 2013). Among various customer segmentation techniques, cluster analysis stands out for its efficacy in managing complex and extensive customer data.

One of the most widely adopted clustering algorithms is the K-means clustering technique. The K-means algorithm partitions data into clusters based on the minimization of within-cluster variance, optimizing the homogeneity of groups formed (MacQueen, 1967). Despite its popularity, traditional K-means clustering faces significant challenges, such as sensitivity to initial cluster center selection and determining the optimal number of clusters. To address this, methods like the elbow method have been introduced, which help identify the appropriate number of clusters by plotting the within-cluster variance against the number of clusters and selecting the "elbow point", where the reduction rate sharply decreases (Thorndike, 1953).

Before clustering, dimensionality reduction is often employed to mitigate noise and

improve computational efficiency. Principal Component Analysis (PCA) is widely used for this purpose, transforming high-dimensional datasets into orthogonal components that retain most of the original variance. PCA facilitates interpretation and allows subsequent clustering to be performed on more stable and condensed representations of customer data (Tsai et al., 2013).

Meanwhile, ensuring the absence of multicollinearity is crucial for model stability and interpretability. The Variance Inflation Factor (VIF) is a diagnostic metric that quantifies how much the variance of an estimated regression coefficient increases due to multicollinearity (James et al., 2013). A high VIF suggests redundant features due to multicollinearity, and such features are often removed to improve model stability and predictive performance (Dupuis and Victoria-Feser, 2013).

Recent work demonstrates the integration of dimensionality reduction and clustering techniques for customer segmentation in e-commerce-related CRM applications (Tsai et al., 2013). Together, PCA, VIF filtering, and K-means clustering are combined to identify distinct customer groups, guiding effective and scalable promotional strategies. This thesis applies this methodology to the company’s customer dataset to extract actionable segments that inform subsequent coupon targeting and outcome modeling.

## 2.4 Modeling Treatment Effects

To personalize promotional offers effectively, it is not enough to predict the likelihood of customer purchase. Marketers must estimate how much more likely a customer is to purchase because they receive a targeted coupon. This is the central aim of uplift modeling, which explicitly estimates the incremental effect of a treatment (e.g., a discount) on an individual’s behavior. Unlike traditional classification models that predict absolute outcomes, uplift models focus on the causal impact of interventions, enabling marketers to target customers for whom the treatment will make the greatest difference (Lo, 2002).

A popular baseline method for modeling uplift is the two-model approach, where separate predictive models are trained for treated and untreated groups. However, this approach may suffer from bias and instability, especially when the data is imbalanced or treatment effects are heterogeneous. To overcome these limitations, researchers have increasingly turned to tree-based ensemble methods, especially Random Forests and Causal Forests, which are more proficient at capturing complex, nonlinear interactions in high-dimensional data (Breiman, 2001; Athey et al., 2019).

Random Forests aggregate predictions from an ensemble of decision trees, each built on a bootstrap sample of the data and a random subset of features. This structure reduces overfitting and improves predictive performance (Breiman, 2001). While originally designed for prediction rather than causal inference, random forests have been adapted for uplift modeling by partitioning the data in ways that highlight differential treatment

effects.

Building on this, Causal Forests extend standard Random Forests to directly estimate Conditional Average Treatment Effects (CATE) at the individual level (Wager and Athey, 2018). They modify tree splitting rules to maximize treatment heterogeneity and employ honest estimation strategies that reduce overfitting, making them particularly suitable for applications such as personalized marketing, where treatment effects vary widely across customer segments.

Complementing uplift estimation, Inverse Probability Weighting (IPW) is a key technique used to adjust for selection bias in observational data. IPW reweights samples according to the inverse of the estimated propensity score (the probability of receiving treatment given covariates) so that treatment and control groups become statistically comparable (Rosenbaum and Rubin, 1983). This technique is especially valuable when the treatment is not randomized, as is often the case in business settings like email marketing campaigns. Studies by Hirano et al. (Hirano et al., 2003) and Austin (Austin and Stuart, 2015) demonstrate that IPW can produce unbiased estimates of average treatment effects when implemented correctly.

Together, causal forests and IPW form a powerful framework for personalized promotion modeling, enabling marketers to identify not just whether a customer will buy, but whether they will buy because of the promotion. This distinction is crucial for allocating limited marketing resources efficiently. Moreover, these methods facilitate policy evaluation by estimating what outcomes would have been under alternative targeting strategies.

In the context of this thesis, these methodologies enable a robust and nuanced evaluation of the company’s email coupon campaigns. By applying causal forests and IPW weighting to historical campaign data, this study seeks to identify optimal discount levels for individual customers, thereby maximizing gain uplift and improving cost-efficiency in promotional targeting.

## 2.5 Evaluating Targeting Strategies Without Deployment

In personalized discount targeting, evaluating alternative promotional strategies through real-world experimentation can be costly, slow, or even infeasible in dynamic business environments. Off-Policy Evaluation (OPE) offers a solution by allowing the estimation of policy performance using existing observational data without testing new strategies in practice (Lemmens et al., 2025). In this thesis, OPE plays a central role in simulating and comparing alternative discount assignment policies based on historical transactional data.

Unlike classical reinforcement learning literature, where doubly robust (DR) and importance sampling (IS) estimators are often required to address bias-variance tradeoffs in

high-dimensional dynamic treatment settings (Thomas, 2016; Jiang and Li, 2016; Dudík et al., 2011), this study adopts a fully model-based approach to off-policy evaluation. Specifically, individualized treatment effects at varying discount levels are estimated using causal forest models (Athey et al., 2019), which directly predict counterfactual outcomes for any hypothetical discount assignment given a customer’s feature profile.

By leveraging model-based estimates, treatment propensities, and robust estimation frameworks, we aim to provide reliable performance predictions for alternative coupon targeting policies, thus enabling data-driven decision-making without the risks of real-world A/B experimentation.

## 3 Methods & Data

### 3.1 Data Description

The dataset used in this thesis originates from the company’s customer relationship management (CRM) system, supplemented by itemized order-level transaction logs. It covers between January 2023 and March 2025, specifically during non-promotional business periods. To reduce confounding effect from major seasonal events, high-intensity promotion periods such as Cyber Week and end-of-season sales were excluded from the sample. Each observation contains the following broad types of information:

1. **Customer features:** Recency, frequency, and monetary (RFM) metrics, web order behavior, historical discount engagement, and customer tier.
2. **Order metadata:** Timestamps, discount status (binary and percentage), and margin and revenue values.

The dataset contains both treatment (discounted vs. non-discounted) and outcome variables (revenue and margin), allowing for causal modeling of treatment effects using uplift estimation methods.

Table 1: Summary of Key Variables

| Variable                 | Description                                                | Type       |
|--------------------------|------------------------------------------------------------|------------|
| L30D_REVENUE             | Revenue from orders in last 30 days                        | Continuous |
| L6M_REVENUE              | Revenue from orders in last 6 months                       | Continuous |
| L12M_REVENUE             | Revenue from orders in last 12 months                      | Continuous |
| L13M_24M_REVENUE         | Revenue from orders in 13 - 24 months window               | Continuous |
| FIRST_REVENUE            | Revenue from customer's first purchase                     | Continuous |
| L30D_FREQUENCY           | Frequency of orders in last 30 days                        | Continuous |
| YTD_FREQUENCY            | Frequency of orders in current year                        | Continuous |
| LIFETIME_FREQUENCY       | Total number of orders during customer lifetime            | Continuous |
| L24M_FULLPRICE_FREQUENCY | Frequency of full-price purchases in past 24 months        | Continuous |
| L25M_36M_WEB_FREQUENCY   | Frequency of online orders in 25 - 36 month window         | Continuous |
| RECENCY                  | Days since last purchase                                   | Continuous |
| EARLY_REPEAT_PURCHASER   | Early repeat purchase indicator (Yes/No)                   | Binary     |
| HAS_PURCHASED_FULL_PRICE | Indicator for full-price purchasing history                | Binary     |
| CUSTOMER_VALUE_TIER_X    | Customer value tiers (Low, Medium, Silver, Gold, Platinum) | Binary     |
| DISCOUNT_PERCENTAGE      | Percentage discount applied to purchase                    | Continuous |
| PURCHASED                | Purchase indicator (Yes/No)                                | Binary     |
| REVENUE                  | Purchase revenue outcome                                   | Continuous |

## 3.2 Data Cleaning and Preparation

### 3.2.1 Handling Missing Values

For categorical indicators, missing values were dropped when extracting the data from company’s database. Missing values appeared in purchase history status typically indicated the absence of past activities. These missing values were filled with zeros to represent no previous activity.

### 3.2.2 Outlier Analysis

To ensure the robustness of the modeling process, outliers were detected and removed using the Isolation Forest algorithm (F. Liu et al., 2008). This unsupervised machine learning technique is suitable for multivariate anomaly detection, as it isolates observations that are substantially different from the majority of the data. The analysis was performed on key revenue and purchase frequency variables.

A contamination rate of 1% was applied, meaning that the algorithm automatically identified the most extreme 1% of data points as outliers based on their multivariate feature profiles. Observations flagged as outliers were excluded from further analysis.

### 3.2.3 Feature Engineering and New Variable Creation

Several new columns were created to enhance model performance and capture customer behavior more accurately:

1. **Purchase Indicator (PURCHASED)**: Binary outcome indicating whether the customer completed a purchase during the observation window.
2. **Month**: Extracted the transaction month from each order date to investigate time-dependent patterns.
3. **Spender Type**: Customers were classified into high spender (highest 10%) and low spender (lowest 10%) groups based on lifetime revenue.

### 3.2.4 Aggregation Levels

The original item-level data was aggregated at both order-level and customer-level for different analyses. Customer-level aggregation enabled customer segmentation. Order-level data allowed examining discount treatment effects at individual purchase level, uplift modeling, causal inference, and policy simulation, combining with segmentation.

### 3.2.5 Exploratory Statistical Tests

Prior to modeling, statistical tests were conducted to investigate whether discount percentages significantly differed across groups:

1. **T-tests:** Two-sample t-tests compared mean discount rates between high spender and low spender groups.
2. **ANOVA:** One-way ANOVA tested whether discount levels varied significantly across months.
3. **Correlation Analysis:** Examined correlations between discount rates and other continuous and binary features (e.g., recency, revenue, customer tier).

Significant differences observed in these tests justified the use of customer segmentation, IPW calculation and causal modeling.

### 3.2.6 Principal Component Analysis (PCA) for Feature Selection

To reduce dimensionality and capture core variance structure, Principal Component Analysis (PCA) was applied to revenue-related and frequency-related features (Jolliffe, 2002). The PCA decomposition is given by:

$$Z = XW \tag{1}$$

where  $X$  is the matrix of standardized original features and  $W$  is the matrix of principal component loadings. Loadings from PCA were analyzed to identify the most influential original features, which were retained for downstream modeling.

Table 2: PCA Loadings for Revenue Features

| Feature          | PC1   | PC2    | PC3    | PC4    | PC5    | PC6    |
|------------------|-------|--------|--------|--------|--------|--------|
| L6M_REVENUE      | 0.416 | -0.231 | -0.125 | -0.275 | -0.009 | -0.184 |
| YTD_REVENUE      | 0.412 | -0.260 | -0.139 | -0.213 | -0.002 | -0.301 |
| L12M_REVENUE     | 0.398 | -0.021 | 0.083  | -0.327 | -0.081 | 0.615  |
| L3M_REVENUE      | 0.380 | -0.312 | -0.057 | 0.171  | 0.035  | -0.332 |
| LIFETIME_REVENUE | 0.361 | 0.243  | 0.447  | -0.109 | -0.022 | 0.279  |
| L30D_REVENUE     | 0.264 | -0.276 | 0.006  | 0.807  | 0.124  | 0.328  |
| L13M_24M_REVENUE | 0.232 | 0.420  | 0.590  | 0.193  | 0.003  | -0.448 |
| LATEST_REVENUE   | 0.230 | 0.459  | -0.467 | 0.205  | -0.689 | -0.020 |
| FIRST_REVENUE    | 0.220 | 0.509  | -0.434 | 0.004  | 0.709  | 0.017  |

Table 3: PCA Loadings for Frequency Features

| Feature                  | PC1   | PC2    | PC3    | PC4    | PC5    | PC6    |
|--------------------------|-------|--------|--------|--------|--------|--------|
| L24M_DISCOUNT_FREQUENCY  | 0.351 | -      | -0.217 | -0.084 | -      | -      |
| LIFETIME_FREQUENCY       | 0.328 | -      | -      | -      | -      | -      |
| L12M_FREQUENCY           | 0.307 | 0.239  | -      | -0.018 | -0.316 | 0.104  |
| L12M_WEB_FREQUENCY       | 0.307 | 0.239  | -      | -0.018 | -0.316 | -      |
| LYR_FREQUENCY            | 0.281 | -      | -0.402 | -0.021 | -0.272 | 0.348  |
| L2YR_FREQUENCY           | 0.262 | -0.322 | -      | 0.017  | 0.148  | -0.197 |
| L13M_24M_FREQUENCY       | 0.262 | -0.303 | -0.274 | 0.015  | 0.308  | -0.178 |
| L13M_24M_WEB_FREQUENCY   | 0.262 | -0.303 | -0.274 | 0.015  | 0.308  | -0.178 |
| L6M_FREQUENCY            | 0.251 | 0.353  | -      | -      | -      | -0.326 |
| YTD_FREQUENCY            | 0.241 | 0.357  | 0.186  | -      | -      | -0.385 |
| L25M_36M_FREQUENCY       | -     | -0.298 | 0.503  | -      | -0.164 | 0.121  |
| L25M_36M_WEB_FREQUENCY   | -     | -0.298 | 0.503  | -      | -0.164 | 0.121  |
| L24M_FULLPRICE_FREQUENCY | -     | -      | -      | 0.992  | -      | -      |
| L3M_FREQUENCY            | -     | 0.318  | 0.169  | 0.048  | 0.326  | -0.332 |
| L30D_FREQUENCY           | -     | -      | 0.116  | 0.069  | 0.569  | 0.683  |

Table 4: PCA Loadings for Customer Value Tier Variables

| Feature                      | PC1    | PC2    | PC3    | PC4    |
|------------------------------|--------|--------|--------|--------|
| CUSTOMER_VALUE_TIER_Low      | 0.875  | 0.295  | -0.084 | -0.171 |
| CUSTOMER_VALUE_TIER_Medium   | -0.453 | 0.778  | -0.127 | -0.211 |
| CUSTOMER_VALUE_TIER_Gold     | -0.131 | -0.472 | -0.688 | -0.341 |
| CUSTOMER_VALUE_TIER_Silver   | -0.100 | -0.275 | 0.706  | -0.470 |
| CUSTOMER_VALUE_TIER_Platinum | -0.049 | -0.099 | 0.073  | 0.767  |

**Feature Selection based on PCA Results.** Following the exploratory PCA analyses for revenue, frequency, and customer value tier variables, the most influential features were selected for inclusion in the modeling pipeline. The loadings from PCA indicated that both recent purchase activities and historical customer characteristics contributed significantly to variance in the dataset. For revenue-related variables, strong contributions to the principal components came from *L12M\_REVENUE*, *L6M\_REVENUE*, *FIRST\_REVENUE*, *L13M\_24M\_REVENUE*, and *L30D\_REVENUE*, reflecting both short-term and longer-term customer spending patterns. Similarly, for frequency-related features, high PCA loadings were observed for *LIFETIME\_FREQUENCY*, *L30D\_FREQUENCY*, *L25M\_36M\_WEB\_FREQUENCY*, *L24M\_FULLPRICE\_FREQUENCY*, and *YTD\_FREQUENCY*, capturing variations in both recent and historical purchasing frequency.



In addition to transactional features, several important customer-level characteristics were retained. These include *RECENCY*; binary indicators *EARLY\_REPEAT\_PURCHASER* and *HAS\_PURCHASED\_FULL\_PRICE*; as well as the full set of customer value tier dummies which summarize historical customer value assessments made by the firm, since four principal components were extracted from five original customer value tier variables, and each component explains a meaningful portion of the variance structure. Finally, a *month* indicator was included to capture potential seasonality and enable control of time-fixed effects during propensity score estimation.

### 3.2.7 Multicollinearity Check with Variance Inflation Factor (VIF)

To ensure that multicollinearity did not compromise model stability, Variance Inflation Factor (VIF) was calculated for the selected features. The VIF is computed as (O’Brien, 2007):

$$VIF_i = \frac{1}{1 - R_i^2} \quad (2)$$

where  $R_i^2$  is the coefficient of determination obtained when regressing feature against all other features.

Table 5: Variance Inflation Factor (VIF) for Selected Features

| Feature                      | VIF  |
|------------------------------|------|
| L12M_REVENUE                 | 7.49 |
| L6M_REVENUE                  | 5.26 |
| FIRST_REVENUE                | 2.48 |
| L13M_24M_REVENUE             | 2.09 |
| L30D_REVENUE                 | 2.40 |
| L25M_36M_WEB_FREQUENCY       | 2.29 |
| L24M_FULLPRICE_FREQUENCY     | 2.30 |
| LIFETIME_FREQUENCY           | 5.82 |
| L30D_FREQUENCY               | 2.59 |
| YTD_FREQUENCY                | 4.43 |
| RECENCY                      | 5.85 |
| EARLY_REPEAT_PURCHASER       | 1.69 |
| HAS_PURCHASED_FULL_PRICE     | 2.33 |
| CUSTOMER_VALUE_TIER_Low      | 3.64 |
| CUSTOMER_VALUE_TIER_Medium   | 1.78 |
| CUSTOMER_VALUE_TIER_Gold     | 1.81 |
| CUSTOMER_VALUE_TIER_Silver   | 1.43 |
| CUSTOMER_VALUE_TIER_Platinum | 3.00 |
| Month                        | 3.68 |

Following PCA-based selection, the retained features were evaluated for multicollinearity using VIF to measure the extent to which each predictor is linearly correlated with other predictors in the model. A commonly accepted threshold for VIF is 10, beyond which multicollinearity may bias model estimates (O’Brien, 2007). As shown in Table 5, all features had VIF values well below 10, suggesting that multicollinearity was not a concern for subsequent model estimation.

### 3.2.8 Standardization

To maintain consistency across scales for PCA, K-means segmentation and IPW calculation, continuous features such as revenue and purchase frequencies were standardized using the Standard Scaler:

$$z = \frac{x - \mu}{\sigma} \quad (3)$$

where  $x$  is the original value,  $\mu$  is the mean of the variable, and  $\sigma$  is the standard deviation.

## 3.3 Analytical Approach

### 3.3.1 Construction of Inferred Negative Samples

One challenge with observational business data is the limited availability of information on customers who did not make purchases. In particular, this dataset lacks full coupon exposure data, which means it is unknown which customers received a discount offer but did not redeem it. To partially address this data limitation, a set of inferred negative samples was constructed using available marketing list data.

Specifically, customers who were included in the email marketing list but made no purchase during the selected observation window were assumed to represent customers who may have received email discount offers but chose not to redeem them. Under this assumption, these non-purchasers are treated as having had potential exposure to coupon offers, but their outcome remains non-purchase ( $Y = 0$ ).

This inferred negative construction allows the model to approximate both successful and unsuccessful coupon redemptions, improving estimation of purchase propensity and treatment effects under observational settings where exposure data are incomplete. The non-buyers’ customer features are still fully incorporated to capture historical behaviors, segment membership, and eligibility for offers.

To avoid introducing imbalances caused by time-varying seasonality and customer heterogeneity, stratified sampling is applied when generating the inferred negatives. Specifically, for each month in the observed transaction data, the number of actual buyers and sampled additional non-buyers from the inactive customer pool followed a 5:1 ratio. That is, for every 100 observed buyers in a given month, 20 inferred non-buyers were sampled. This ensured that the temporal distribution of the inferred negatives matched

the seasonality of observed buyers. The sampled non-buyers were also assigned a corresponding “month” variable to align with the buyer groups they were sampled against. This allowed the month-level time effects to be included later in the uplift modeling and policy evaluation stages.

For the inferred negatives, feature engineering steps such as recency, frequency, and customer tier encoding were still applied based on available historical customer data. The revenue, margin, and purchase outcome were set to zero, and treatment was by default set to  $T = 0$  before propensity score estimation and inverse probability weighting.

This construction allowed for approximating a richer set of counterfactuals, helping the causal forest and propensity score models better distinguish between treated and untreated units, purchased and not purchased, under the assumption of unconfoundedness conditional on observables.

### 3.3.2 Customer Segmentation using K-Means Clustering

Customer segmentation involved clustering customers based on standardized features (revenue, recency, and frequency) and other binary indicators (customer tier, early repeat purchaser, and full price purchaser). K-means clustering method (MacQueen, 1967) was used, defined mathematically as:

$$\arg \min_C \sum_{i=1}^k \sum_{x \in C_i} |x - \mu_i|^2 \quad (4)$$

where  $\mu$  represents the mean of points in cluster  $i$ , and  $k$  is the number of clusters.

To select the optimal number of clusters, the Elbow method was applied. This involved plotting within-cluster variance (inertia) against the number of clusters. The number of clusters was chosen at the point where adding an additional cluster no longer significantly reduced inertia.

### 3.3.3 Propensity Score Estimation and Inverse Probability Weighting (IPW)

To address selection bias, logistic regression was employed to estimate propensity scores following Rosenbaum and Rubin’s framework (Rosenbaum and Rubin, 1983), defined as the probability of a customer receiving a discount given observed characteristics :

$$\text{logit}(p(X)) = \ln \left( \frac{p(X)}{1 - p(X)} \right) = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n \quad (5)$$

To control for potential seasonality and time-varying treatment assignment probabilities, a time fixed effect was included in the propensity score model by including the transaction month as an additional covariate. Discount policies and customer behaviors may shift across months due to marketing campaigns, promotions, or external seasonality (e.g., holiday periods). And the ANOVA test shows that there is a statistically

significant difference in discount rates across different months. Including the month indicator ensures that propensity score estimation remains unbiased and properly reflects the business process that affect discount assignment over time.

Logistic regression was selected to estimate propensity scores because it provides direct probability outputs and straightforward interpretability for treatment assignment modeling. More complex machine learning models such as random forests or gradient boosting could have been applied, potentially offering better predictive performance. However, these models may introduce overfitting and reduce transparency, especially when estimating inverse probability weights (IPW) where stability and overlap are critical. Logistic regression ensures stable PS estimation, which is particularly important given the presence of inferred negatives in the data construction.

Using these propensity scores, Inverse Probability Weighting (IPW) adjusted for selection bias. The IPW for each individual was computed as:

$$IPW_i = \frac{T_i}{p(X_i)} + \frac{1 - T_i}{1 - p(X_i)} \quad (6)$$

where  $T$  indicates treatment (receiving a discount).

IPW was selected to address treatment assignment imbalance while preserving the full observational dataset. Although doubly robust methods like targeted maximum likelihood estimation (TMLE) or covariate balancing propensity score (CBPS) offer alternative balancing strategies, IPW was chosen for its simplicity, transparency, and wide adoption in uplift modeling and causal forest pipelines (Imai and Ratkovic, 2013).

The Receiver Operating Characteristic (ROC) curve was used to evaluate the performance of the propensity score model (Fawcett, 2006), as shown in Figure 1. The optimal classification threshold 0.376 was selected based on the Youden Index (Youden, 1950), which maximizes the sum of sensitivity ( $TruePositives / (TruePositives + FalseNegatives)$ ) and specificity ( $TrueNegatives / (TrueNegatives + FalsePositives)$ ). This threshold reflects the estimated likelihood above which customers are predicted to have received discount treatment based on their observable features.

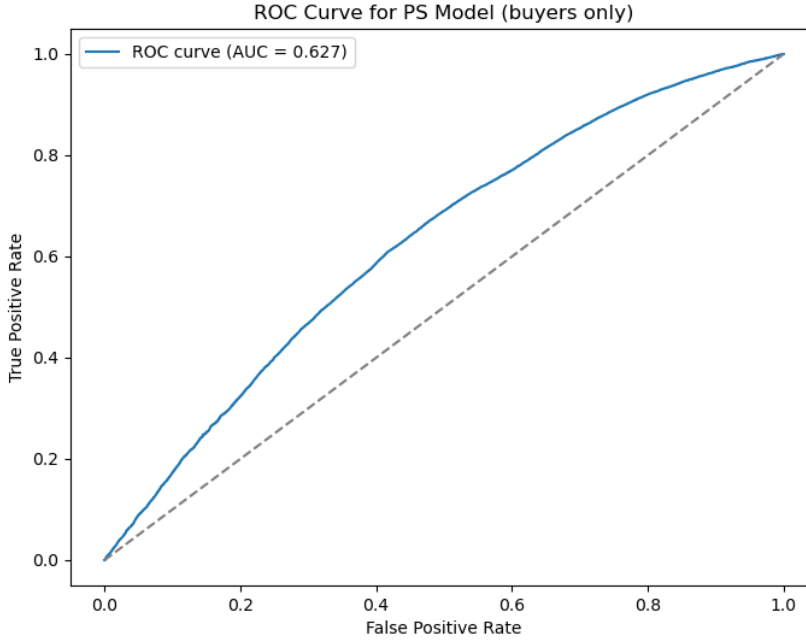


Figure 1: ROC Curve for Propensity Score Model.

It is important to note that while the full dataset for causal modeling included both observed buyers and inferred negatives, the propensity score model itself was trained exclusively on the observed buyer group. This was done to prevent data leakage, ensure proper model calibration, and preserve the causal interpretability of treatment assignment. The inferred negatives were excluded from model fitting to avoid introducing biases in estimating the treatment assignment mechanism.

After the model was trained, propensity scores were predicted for both buyers and inferred negatives. For the observed buyers, the true treatment (discount usage) was directly observed. For inferred negatives, the predicted propensity scores were further used to assign treatment labels. Specifically, inferred negatives with predicted propensity scores above the selected threshold were assumed to have likely received discount offers (assigned  $T = 1$ ), while those below the threshold were assumed untreated ( $T = 0$ ). For those treated customers, the discount percentage was assigned to 15 since it is the most common discount rate across all customer group and all months. This approach allows partially recovering potential coupon exposure states for non-purchasing customers where actual exposure information was unavailable.

Inverse probability weights (IPW) were then calculated for both buyers and inferred negatives based on these estimated treatment assignments, allowing the full sample to be reweighted for balanced covariate distributions during subsequent causal modeling.

### 3.3.4 Causal Forest Modeling for Uplift Estimation

We employed Causal Forest models to estimate heterogeneous treatment effects (uplift) across different customers. Causal Forest uses decision trees to estimate conditional average treatment effects (CATE):

$$\tau(x) = \mathbb{E}[Y(1) - Y(0)|X = x] \quad (7)$$

Random Forest regressors served as the base learners because of their flexibility and ability to handle non-linear relationships effectively. The Causal Forest algorithm handles potential overfitting through cross-validation and regularization techniques.

Causal Forest (Athey et al., 2019) was selected for treatment effect estimation because it directly models heterogeneous treatment effects while accounting for covariate interactions. Alternative methods such as uplift trees, meta-learners (S-learner, T-learner, X-learner), and Bayesian Additive Regression Trees (BART) exist. Uplift trees may suffer from high variance with limited data. Meta-learners require careful model selection for both outcome and treatment models and can struggle with stability when treatment assignment is highly imbalanced. BART offers flexible nonparametric estimation but is computationally intensive for large-scale evaluation. Causal Forest provides an efficient compromise, producing well-adjusted CATE estimates while scaling effectively to large customer datasets.

To capture trade-offs between sales volume and revenue, two causal forest models were trained with different outcome variables:

1.  $Y^{(purchase)}$ : Binary purchase outcome;
2.  $Y^{(revenue)}$ : Observed revenue amount per customer.

For each outcome, corresponding Causal Forest models (Athey et al., 2019) were trained incorporating IPW weights derived from the earlier propensity score model. Using separate models allowed isolating both volume-based and profit-based policy performance metrics under simulated counterfactual policies.

### 3.3.5 Off-Policy Evaluation

The most crucial component of this study involves developing an individualized discount policy that balances purchase uplift against discount cost to maximize business margin. To achieve this, a multi-stage policy design combining both segment-level and individual-level estimation was constructed.

**Segment-level optimization via off-policy evaluation (OPE).** First, segment-level optimal discount rates were estimated using off-policy evaluation (OPE). Customers were first grouped by segment (obtained from K-means clustering) and by transaction

month. Within each segment-month group, expected purchases and revenues uplift under different discount levels were simulated using the fitted causal forest models.

The OPE logic follows:

$$V(d) = \mathbb{E} [\hat{Y}(T = d, X)], \quad (8)$$

where  $V(d)$  is the expected revenue or purchase outcome at discount level  $d$ , and  $\hat{Y}(T = d, X)$  is predicted using the random forest model for each discount level applied to feature set  $X$ . The segment-level optimal discount was selected as the discount that maximized expected gain within each segment:

$$\text{Gain} = (\text{Revenue} - \text{Discount Amount}) \times \text{Purchase}, \quad (9)$$

where the discount amount is reconstructed as:

$$\text{Discount Amount} = \frac{\hat{Y}(T = d, X)}{1 - d/100} \times \frac{d}{100}. \quad (10)$$

This segment-level simulation enabled the model to approximate counterfactual business outcomes for all possible discount levels within each customer segment.

**Individual-Level Treatment Assignment and Threshold Tuning.** While the segment-level optimal discount policy served as a starting point, additional individualized refinement was implemented to prevent unnecessary discount allocation to customers unlikely to respond. The final treatment assignment rule was developed based on a two-stage thresholding process, combining both predicted purchase probabilities and estimated uplift effects from the causal forest model.

In the first stage, a grid search procedure was conducted to tune the purchase propensity threshold. For each customer, the baseline purchase probability without discount was predicted using the trained purchase propensity model. A range of candidate thresholds (from 0.45 to 0.98, incremented by 0.05) was evaluated. Customers whose predicted purchase probabilities under segment-level discounts did not exceed the threshold were excluded from assigning segment-level discounts. This initial screening aimed to restrict discounts to customers with sufficiently high predicted likelihood of conversion.

In the second stage, an uplift-based filtering layer was applied. For each customer, the individualized uplift effect was estimated from the fitted causal forest model under the assigned segment-level discount. A separate grid search was performed over uplift thresholds ranging from  $-0.1$  to  $0.1$  (incremented by  $0.01$ ). For each uplift threshold candidate, only customers whose estimated uplift exceeded the threshold were finally assigned the segment discount; otherwise, treatment was denied. This second layer ensured that discounts were allocated only to customers who demonstrated positive and meaningful incremental response to discount exposure.

The complete policy rule is summarized as:

$$T_i = \begin{cases} \text{Segment Optimal Discount,} & \text{if } \hat{P}_i > \lambda_p \text{ and } \hat{\tau}_i > \lambda_\tau \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

For each combination of purchase probability threshold and uplift threshold, expected purchase, revenue, gain, and total discount cost were simulated.

The logic behind these dual thresholds is both theoretical and practical. High purchase probability ensures that discounts are not wasted on unlikely buyers. Positive uplift ensures that discounts are only applied when they genuinely increase the likelihood of purchase compared to no treatment. This design minimizes unnecessary discounting while focusing resources on highly responsive customers.

**Elasticity Test by Segment** To further evaluate the responsiveness of different customer segments to discount changes, an elasticity test was conducted. The goal of this analysis was to investigate whether an increase in discount levels leads to a substantial shift in the number of predicted purchasers across segments.

Within each segment, predicted purchase probabilities were obtained using the trained causal forest model under varying discount levels. Customers with predicted purchase probability above the purchase threshold were classified as *predicted purchasers*, while those below were classified as *predicted non-purchasers*.

Discount levels were simulated across eight values: 0%, 10%, 15%, 20%, 25%, 30%, 40%, and 50%. For each combination of segment and discount level, the counts of predicted purchasers and non-purchasers were recorded.

**Bootstrap sensitivity evaluation for policy stability.** To assess the robustness of the policy recommendations, bootstrap resampling was applied to the evaluation sample (Tibshirani and Efron, 1993). This generated empirical confidence intervals around the expected purchase uplift, providing additional validation of stability across random customer samples.

**Model Reliability Check via Recall Estimation** In addition to bootstrap analysis for uplift stability, a model reliability check is conducted to assess whether the model can correctly predict historical purchase outcomes under the actually assigned discount levels. This step helps evaluate whether the model successfully captures purchase behavior under observed business policies.

For each customer in the test set, their historically observed discount percentage is assigned as input to the trained causal forest model. The model then predicted purchase probabilities under these actual discounts. With the purchase threshold applied to convert



predicted probabilities into binary purchase outcomes, the predicted labels were then compared to the actual observed outcomes.

Model recall was calculated as:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

Where true positives refer to correctly predicted purchasers, and false negatives refer to actual purchasers that the model predicted as non purchasers. This recall metric serves as a diagnostic for the model’s sensitivity in recognizing purchase behaviors under business-as-usual discount assignments.

## 4 Results

### 4.1 Exploratory Analysis of Treatment Variation

Before developing predictive models, descriptive statistical analyses were conducted to examine how discount treatment varied across different customer groups and time periods. This allowed evaluating whether discount allocation was systematically related to customer characteristics.

#### 4.1.1 Distribution of Discount Rate by Spender Level

As mentioned, customers were first categorized into *high spenders* and *low spenders* based on lifetime purchase revenue. Figure 2 shows the distribution of discount percentages for the two spender groups.

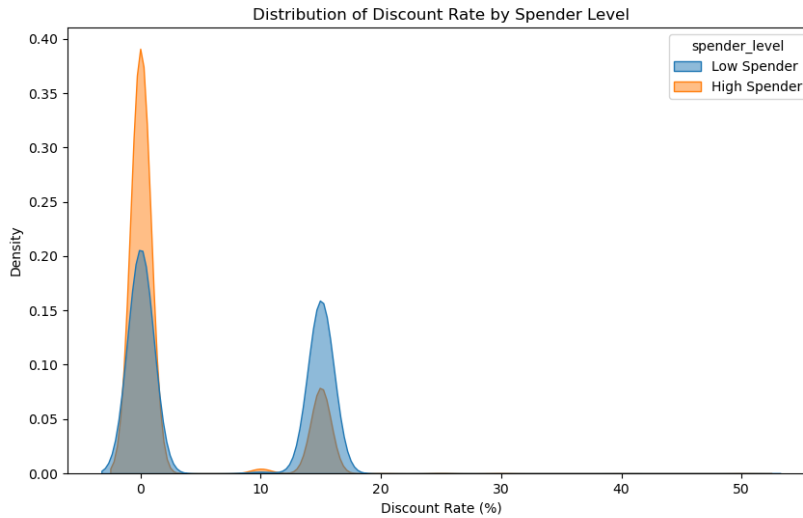


Figure 2: Distribution of discount rate by spender level.

The plot shows that both spender groups receive discounts, but the density differs across groups. Low spenders tend to receive slightly higher discount rates, with more density in the 10–20% range. High spenders receive slightly fewer high discounts, suggesting a degree of business targeting already present in the historical data.

#### 4.1.2 Correlation Between Treatment and Customer Features

To further investigate whether discount assignment depends on customer characteristics, several statistical tests were conducted:

**T-test for Spender Groups.** Two-sample t-tests compared the mean discount rates between high spenders and low spenders. The result shows a statistically significant difference ( $p < 0.001$ ), confirming that low spenders on average received higher discounts than high spenders.

**ANOVA for Time Effect.** A one-way ANOVA was performed to test whether discount rates varied significantly across different months. The test yielded an  $F$ -statistic of 23.74 with a  $p$ -value  $< 0.001$ , indicating statistically significant variation in discount assignment over time, likely reflecting seasonal marketing campaigns.

**Spearman Correlation with Customer Features.** Spearman rank correlation coefficients were computed between discount percentage and continuous customer features such as revenue, frequency, recency, and order history. Table 6 summarizes the results.

Table 6: Spearman Correlation between Features and Discount Percentage

| Feature                      | Spearman<br>Correlation | P-Value  |
|------------------------------|-------------------------|----------|
| LIFETIME_FREQUENCY           | −0.224                  | < 0.0001 |
| L12M_REVENUE                 | −0.132                  | < 0.0001 |
| L25M_36M_WEB_FREQUENCY       | −0.119                  | < 0.0001 |
| L13M_24M_REVENUE             | −0.118                  | < 0.0001 |
| CUSTOMER_VALUE_TIER.Low      | 0.104                   | < 0.0001 |
| EARLY_REPEAT_PURCHASER       | −0.103                  | < 0.0001 |
| YTD_FREQUENCY                | −0.082                  | < 0.0001 |
| RECENCY                      | 0.080                   | < 0.0001 |
| CUSTOMER_VALUE_TIER.Gold     | −0.079                  | < 0.0001 |
| L6M_REVENUE                  | −0.078                  | < 0.0001 |
| CUSTOMER_VALUE_TIER.Platinum | −0.076                  | < 0.0001 |
| L30D_REVENUE                 | −0.058                  | < 0.0001 |
| L30D_FREQUENCY               | −0.058                  | < 0.0001 |
| CUSTOMER_VALUE_TIER.Silver   | −0.042                  | < 0.0001 |
| CUSTOMER_VALUE_TIER.Medium   | 0.029                   | < 0.0001 |
| L24M_FULLPRICE_FREQUENCY     | 0.023                   | < 0.0001 |
| HAS_PURCHASED_FULL_PRICE     | 0.021                   | < 0.0001 |
| FIRST_REVENUE                | −0.020                  | < 0.0001 |

These results suggest that customers with lower spending and lower engagement tend to receive higher discounts. This evidence motivates the need to control for these confounding variables when estimating causal treatment effects later.

## 4.2 Customer Segmentation Results

Following data cleaning, standardization, and feature selection, K-means clustering was applied to identify customer segments based on their behavioral and purchase patterns. The optimal number of clusters was determined using the Elbow Method. Figure 3 displays the within-cluster sum of squares (WCSS) for different cluster numbers.

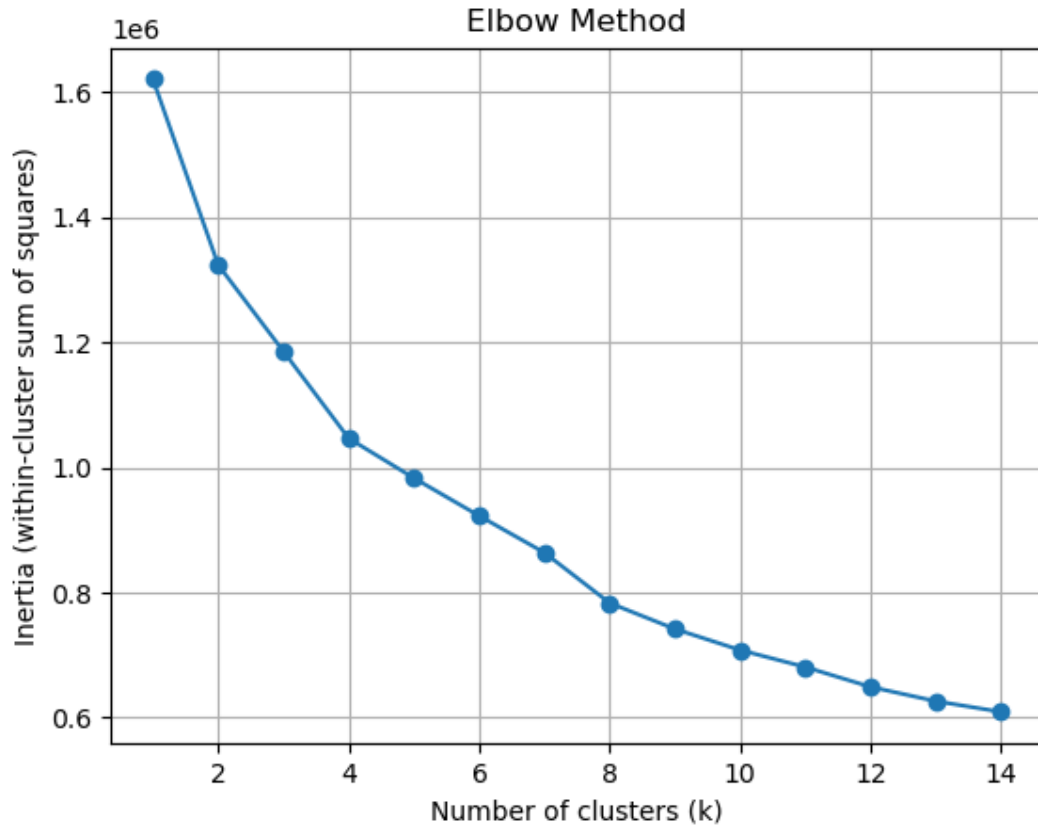


Figure 3: Elbow method for determining optimal number of clusters.

Based on the elbow point observed at four clusters,  $k = 4$  was selected as the final number of clusters. This choice reflects a balance between model simplicity and capturing sufficient heterogeneity in customer behavior.

Table 7 summarizes the key characteristics of each segment based on the main revenue, frequency, recency, and customer tier features.

Table 7: Summary Statistics by Customer Segment

| Feature                  | Segment | Segment | Segment | Segment |
|--------------------------|---------|---------|---------|---------|
|                          | 1       | 2       | 3       | 4       |
| L12M_REVENUE             | ■       | ■       | ■       | ■       |
| L6M_REVENUE              | ■       | ■       | ■       | ■       |
| FIRST_REVENUE            | ■       | ■       | ■       | ■       |
| L13M_24M_REVENUE         | ■       | ■       | ■       | ■       |
| L30D_REVENUE             | ■       | ■       | ■       | ■       |
| L25M_36M_WEB_FREQUENCY   | ■       | ■       | ■       | ■       |
| L24M_FULLPRICE_FREQUENCY | ■       | ■       | ■       | ■       |
| LIFETIME_FREQUENCY       | ■       | ■       | ■       | ■       |
| L30D_FREQUENCY           | ■       | ■       | ■       | ■       |
| YTD_FREQUENCY            | ■       | ■       | ■       | ■       |
| Recency                  | ■       | ■       | ■       | ■       |
| Customer Count           | ■       | ■       | ■       | ■       |

Based on the summary statistics, clear behavioral patterns emerge across customer segments. The key differentiating features include recent and historical revenue, recent and historical purchase frequency, and recency. Segment 2 appears to represent inactive or at-risk customers with low recent revenue and very high recency, while Segment 1 contains moderately active customers with moderate revenue and frequency. Segment 3 includes historically valuable customers with high purchase frequencies and strong historical revenue, while Segment 4 reflects highly active recent purchasers with high recent revenue and very low recency. These patterns provide managerial insights that align with common RFM-based segmentation logic and help validate the customer clustering structure.

### 4.3 Uplift Modeling and Off-Policy Evaluation Results

#### 4.3.1 Causal Forest Uplift Estimation

After constructing the balanced dataset with observed buyers and inferred negatives, the Causal Forest model was employed to estimate individualized treatment effects across customers. Separate models were trained for both purchase propensity and revenue outcome variables, allowing the estimation of heterogeneous responses to discounts.

For both models, hyperparameter tuning was applied, including number of trees (2000 estimators), minimum leaf size (200), and cross-validation folds (4-fold CV). The final models were trained using inverse probability weights (IPW) estimated from the propensity score model as sample weights, ensuring balanced covariates across treatment condi-

tions. This setup allows for unbiased estimation of conditional average treatment effects (CATE).

### 4.3.2 Threshold Calibration for Treatment Assignment

Following the estimation of individualized treatment effects through the causal forest model, a two-step threshold calibration was conducted to determine the optimal assignment rule for discount treatments. The goal of this calibration was to identify customers who were both highly likely to convert after receiving a discount and for whom the incremental effect (uplift) of discounting would justify the treatment cost.

The purchase probability threshold was tuned first by evaluating performance across a grid of thresholds ranging from 0.45 to 0.98. As shown in Figure 4, total gain remained relatively stable across most values until the purchase probability threshold reached 0.90, from which point it showed an uptrend. Therefore, a purchase probability threshold of 0.90 was selected.

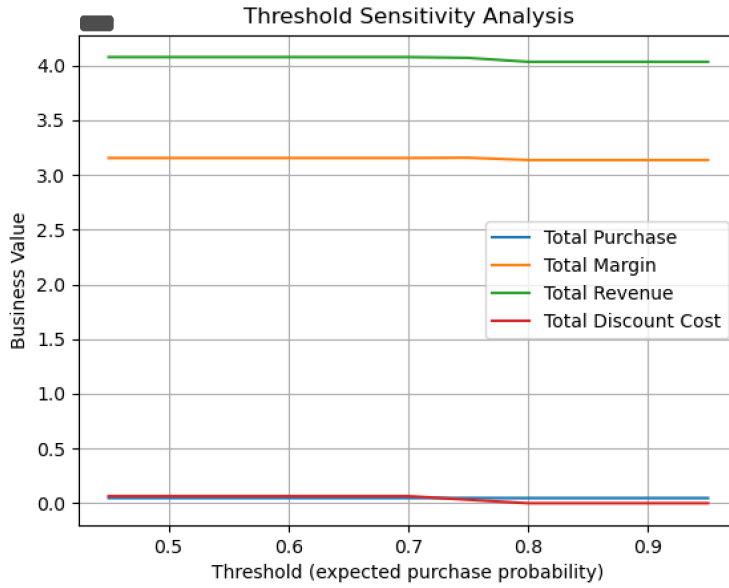


Figure 4: Threshold Sensitivity Analysis for Predicted Purchase Probability.

Next, the uplift threshold was tuned using a fine grid from -0.1 to 0.1. As shown in Figure 5, the revenue remained stable as the uplift threshold moved toward zero. However, as the uplift threshold increased from 0 to 0.01, even a small uplift threshold of 0.01 resulted in substantial cost savings while preserving nearly all revenue improvements, and the total discount costs sharply decreased without harming the gain in revenue, reflecting improved targeting precision in excluding non-responsive customers.

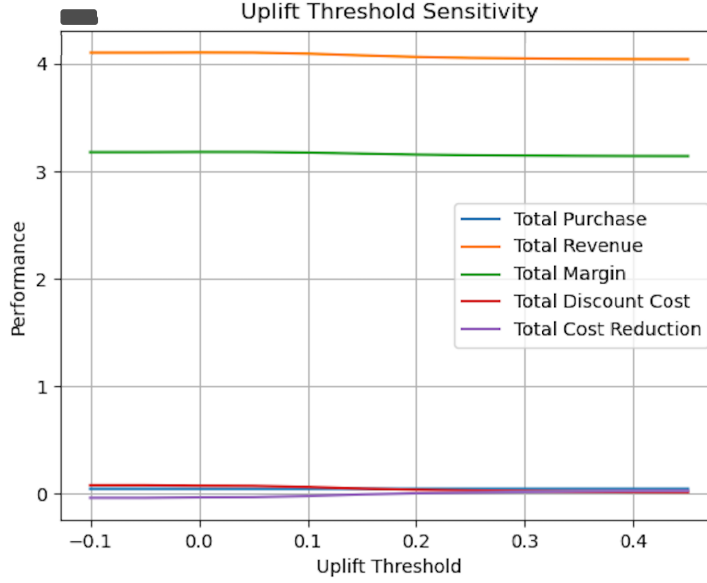


Figure 5: Threshold Sensitivity Analysis for Estimated Uplift Effect.

The final decision rule assigns a discount only when both conditions are met simultaneously:

$$\hat{P}_i \geq 0.90 \quad \text{and} \quad \hat{\tau}(X_i) \geq 0.01 \quad (12)$$

This dual-threshold rule balanced between maximizing profit uplift and minimizing unnecessary discount costs, by selectively targeting customers who are both highly convertible and responsive to discount incentives.

#### 4.3.3 Off-Policy Evaluation (OPE)

The off-policy evaluation was conducted to assess the performance of the proposed profit-optimized policy relative to the historical actual policy. The results are summarized in Table 8.

Table 8: Off-Policy Evaluation Results

| Metric                               | Actual Policy | Profit-Optimized Policy |
|--------------------------------------|---------------|-------------------------|
| Expected Purchase Probability        | ■             | ■                       |
| Expected Revenue                     | ■             | ■                       |
| Gain (Revenue – Discount) × Purchase | ■             | ■                       |
| Policy Gain for Purchase             |               | 0.0055                  |
| Policy Gain for Revenue              |               | 0.2787                  |

The optimized policy achieved an additional 0.0055 increase in expected purchase probability, an increase of 0.2787 in expected revenue, and a significant improvement in expected gain (profit), which increased by 4.8.

Although the average gain in purchase probability and revenue appears relatively small in absolute terms, even marginal improvements in these metrics can translate into substantial business impact when scaled across a large customer base. More importantly, the profit gain highlights that the optimized policy successfully identifies customers who are most likely to purchase with lower levels of discount incentives, allowing the firm to achieve higher profitability without sacrificing sales volume. The result suggests that more targeted discount allocation can simultaneously improve efficiency (less unnecessary discounting) and maintain revenue gain.

#### 4.3.4 Discount Cost Reduction

One of the most significant benefits of the optimized policy was the substantial reduction in total discount expenditures. Table 9 summarizes the discount amounts under both policies.

Table 9: Discount Cost Comparison

| Metric                | Actual Policy | Optimized Policy |
|-----------------------|---------------|------------------|
| Total Discount Amount | ■             | ■                |
| Discount Saving       |               | ■                |

Although the optimized policy achieved only marginal improvements in purchase probability and revenue, it resulted in a very large reduction in discount spending. This highlights that by carefully assigning discounts only where they are likely to create incremental lift, the firm can preserve revenue while dramatically lowering overall costs.

#### 4.3.5 Elasticity Test by Segment

Table 10 summarizes the results of the elasticity test. The results show heterogeneous discount elasticity across segments.

In Segment 1, representing moderately active customers with mid-level spending, there was a gradual increase in predicted purchasers as discounts rose, though the effect flattened beyond 30%, suggesting diminishing returns at higher discount levels. Segment 3, which includes many high-value and historically loyal customers, showed limited sensitivity to additional discounting. Most customers in this group were already likely to purchase even without discounts, and higher discounts only converted a small additional portion of non-purchasers. Segment 2, identified as inactive or at-risk customers, displayed minimal responsiveness to discounts. The number of predicted purchasers even declined as discounts increased, indicating that discounting may not be an effective strategy for reactivating this group. In contrast, Segment 4 exhibited a stronger and more continuous response to discounting, with non-purchasers steadily converting into purchasers as dis-



counts increased. This suggests that Segment 4 customers are more responsive to pricing incentives and may benefit most from targeted discount strategies.

Overall, the elasticity analysis confirms heterogeneous discount responsiveness across segments and supports the implementation of a dual-threshold policy that assigns discounts more selectively, rather than applying broad discount increases.

Table 10: Elasticity Test: Purchaser and Non-Purchaser Counts by Segment and Discount Level

| Segment | Non-Purchasers | Purchasers | Discount (%) |
|---------|----------------|------------|--------------|
| 1       | ■              | ■          | 0            |
| 1       | ■              | ■          | 10           |
| 1       | ■              | ■          | 15           |
| 1       | ■              | ■          | 20           |
| 1       | ■              | ■          | 25           |
| 1       | ■              | ■          | 30           |
| 1       | ■              | ■          | 40           |
| 1       | ■              | ■          | 50           |
| 2       | ■              | ■          | 0            |
| 2       | ■              | ■          | 10           |
| 2       | ■              | ■          | 15           |
| 2       | ■              | ■          | 20           |
| 2       | ■              | ■          | 25           |
| 2       | ■              | ■          | 30           |
| 2       | ■              | ■          | 40           |
| 2       | ■              | ■          | 50           |
| 3       | ■              | ■          | 0            |
| 3       | ■              | ■          | 10           |
| 3       | ■              | ■          | 15           |
| 3       | ■              | ■          | 20           |
| 3       | ■              | ■          | 25           |
| 3       | ■              | ■          | 30           |
| 3       | ■              | ■          | 40           |
| 3       | ■              | ■          | 50           |
| 4       | ■              | ■          | 0            |
| 4       | ■              | ■          | 10           |
| 4       | ■              | ■          | 15           |
| 4       | ■              | ■          | 20           |
| 4       | ■              | ■          | 25           |
| 4       | ■              | ■          | 30           |
| 4       | ■              | ■          | 40           |
| 4       | ■              | ■          | 50           |

#### 4.3.6 Robustness Check via Bootstrapping

To assess the robustness and stability of the estimated uplift effects, a bootstrap simulation was conducted. In total, 500 bootstrap samples were drawn with replacement from the test dataset, and the policy evaluation procedure was repeated for each sample to estimate variability in the uplift effect.

The results of the bootstrap analysis are summarized in Table 11.

Table 11: Bootstrap Robustness Check for Uplift Effect (500 Samples)

| Statistic               | Value            |
|-------------------------|------------------|
| Mean uplift (bootstrap) | 0.0048           |
| Standard deviation      | 0.0003           |
| 95% Confidence Interval | [0.0042, 0.0053] |

As shown in Table 11, the uplift effect remained highly stable across repeated samples, with only minor variation. The 95% confidence interval indicates that the expected uplift falls between 0.0042 and 0.0053. This range is very narrow, suggesting that the estimated policy gain is robust to random fluctuations in the customer sample.

The corresponding distribution of uplift estimates from the bootstrap simulations is visualized in Figure 6.

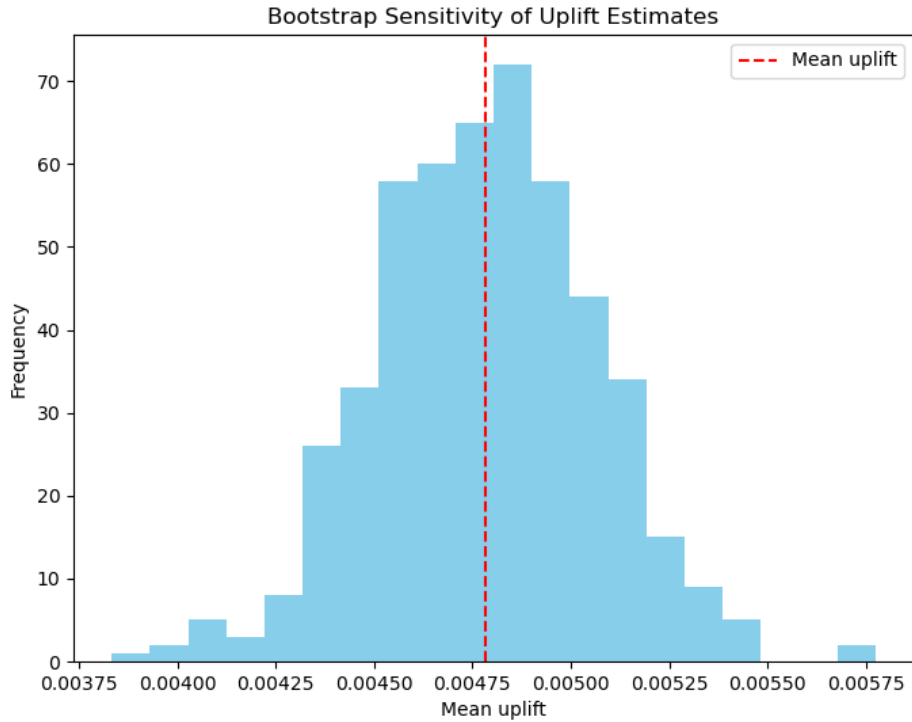


Figure 6: Distribution of Uplift Estimates from 500 Bootstrap Samples

This robustness check provides additional confidence in the stability and practical applicability of the optimized discount policy derived from the uplift model.

#### **4.3.7 Model Reliability under Historical Discounts**

To further validate model performance, whether the uplift model can correctly reproduce historical purchase outcomes under the actual discount levels assigned by the business is tested.

For each customer, the model predicted purchase probability using their actual assigned discount. After applying a classification threshold of 0.9, model predictions were compared with observed purchase outcomes. The recall score achieved was 0.9813, indicating that the model was able to correctly identify approximately 98% of observed purchases.

This result demonstrates that the model is highly capable of capturing customer purchase responses to historical discounts, providing confidence that its predictions under counterfactual discount assignments are reasonably reliable.

## **5 Conclusion and Discussion**

### **5.1 Summary of Key Findings**

The purpose of this research was to develop an optimized discount allocation policy through uplift modeling and causal inference, using transaction data from the company’s email marketing activities. The study covered a two-year period from January 2023 to March 2025. It integrated advanced modeling steps such as data preprocessing, feature selection through principal component analysis, propensity score estimation, inverse probability weighting, causal forest modeling, off-policy evaluation, and profit-based optimization.

The model successfully estimated the individualized effect of discounts on both purchase probability and revenue outcomes. Under the optimized discount assignment, the expected purchase probability increased by 0.0055 compared to the actual policy. In addition, expected revenue increased by 0.2787. Most importantly, the optimized strategy led to a significant improvement in profit gain, which was defined as revenue minus discount amount multiplied by purchase probability. Gain increased by 4.8 under the optimized policy. At the same time, the total discount cost was reduced by 89.6%.

An elasticity test was also conducted to investigate the responsiveness of customers to different discount levels, and it showed that some segments are less sensitive to discount changes, which suggests the need for differentiated discount strategies.

Throughout the modeling process, model reliability was tested through several robustness checks. The recall test showed that the model correctly predicted approximately

98% of historical purchases under actual discounts. Bootstrap tests further confirmed the stability of the uplift estimates.

## 5.2 Key Contributions

The research contributes both methodologically and practically to the field of personalized promotion optimization. Methodologically, it demonstrates how uplift modeling combined with causal inference can estimate counterfactual outcomes using observational data. This is valuable in business environments where randomized experiments are difficult to conduct. The combination of causal forests and off-policy evaluation provides a flexible and robust way to estimate treatment effects in non-experimental settings.

From a practical perspective, the study introduces a rule-based segmentation approach that managers can implement based on the K-means clustering results and feature profiles. Segments can be created using simple and interpretable features such as revenue, frequency, recency, and historical value tiers. Managers can then train models on these manually defined segments to improve business control and interpretability.

The study also offers a strategic contribution by showing that profit gains were driven not only by higher revenue but also by reduced discount costs. Since the revenue values are observed directly from transaction data, the increase in profit reflects mainly the result of assigning discounts more efficiently and avoiding over-discounting. The freed-up discount budget can potentially be reallocated toward reaching broader customer groups, including less active or inactive consumers who might be converted into stable buyers through email marketing or social media advertising.

## 5.3 Strategic Recommendations

The research provides several actionable recommendations for managers who want to apply individualized discount targeting in real business settings. While the modeling in this study demonstrates the potential of highly personalized treatment assignment, full individual-level targeting may not always be feasible in practice due to operational constraints in email marketing systems. As an alternative, managers can adopt a rule-based segmentation approach that integrates insights from the analysis while remaining easy to implement.

For example, when planning a win-back campaign, managers may first apply simple selection rules to define an initial target group, such as customers who have not made any purchase in the past 12 months. This initial filtering step ensures that discount offers are focusing on customers who are at risk of churn and may benefit most from targeted promotions.

Within this selected population, more granular segmentation can then be applied by combining with additional features such as historical revenue, purchase frequency,

and customer value tier. The resulting sub-segments allow managers to profile customer groups with different purchase behaviors and potential responsiveness. Uplift modeling results can inform the expected treatment effects for each sub-segment. The dual-threshold rule combining predicted purchase probability and uplift estimates provides a simple yet effective decision framework: sub-segments with high purchase probability but low uplift may not require discounting, while sub-segments with very low purchase probability may be unlikely to convert even with discounts. By excluding these sub-segments, managers can focus discount allocation on the sub-segments that present the greatest opportunity for incremental profit gain.

This rule-based hybrid approach serves as a more operationally feasible substitute for full individualized assignment, balancing business constraints with data-driven targeting insights. While the results in this study were obtained using observational data and off-policy evaluation, managers are encouraged to validate these recommendations through small-scale A/B tests before fully implementing the recommended policy, to validate its effects in real campaigns if needed.

## 5.4 Limitations and Future Research

Several limitations should be acknowledged. The data used in this study lacked direct information on coupon exposure for each customer. Due to this limitation, inferred negatives were constructed from non-buyers in the email marketing list whose purchase behavior during the observation window was zero. Treatment assignments for these inferred negatives were estimated indirectly based on predicted propensity scores. Having direct coupon exposure information would allow for more precise treatment labeling and potentially improve the accuracy of the treatment effect estimates.

The dataset also lacked detailed time stamps on coupon redemption. This limited the ability to study time dynamics, such as whether customers redeemed coupons immediately or after a delay. Future research could benefit from incorporating time-to-event analysis if such data become available.

Although causal forests were used as the main uplift modeling method, future research could experiment with alternative algorithms such as meta-learners, T-learners, or X-learners, to compare their performance in continuous treatment effect estimation.

Lastly, while the current model optimized individual-level profit gain, it did not directly consider overall campaign budget constraints or promotion spend limits. Future extensions may take such business constraints into the optimization process to support full-scale operational planning.

## 5.5 Reflection

The findings of this thesis highlight both the opportunities and challenges in optimizing discount allocation through data-driven personalization. While the uplift modeling framework successfully identifies segments that respond differently to discount levels, the results suggest that not all customer groups benefit equally from discount interventions. In particular, for high-value customers (Segment 4), targeted discounts can effectively stimulate additional purchases with minimal cost inefficiency. However, for larger segments such as inactive or moderately active customers (Segments 2 and 1), the elasticity analysis reveals more complex dynamics. In Segment 2, increasing discount levels did not generate additional conversions but instead correlated with decreasing predicted purchase rates, suggesting that disengagement in this group may stem from factors beyond price sensitivity.

These patterns raise important managerial considerations. While discounts remain an effective lever for certain responsive segments, relying solely on discounting for broader customer groups risks both inefficient resource allocation and potential brand erosion. Repeated heavy discounting may lead even loyal customers to delay purchases in hope of future promotions, undermining brand equity and long-term profitability.

To mitigate these risks, this thesis proposed a dual-threshold strategy, and the proposed data-driven framework assigns only the minimum discount necessary to trigger meaningful uplift, balancing conversion gains with cost efficiency. For segments demonstrating low price elasticity, non-price-based marketing interventions may offer a more sustainable way to re-engagement. Brand-building activities such as social media campaigns, influencer partnerships, and offline visibility through billboards or athlete sponsorships can reinforce brand perception and stimulate purchase intent among broader audiences without over-reliance on promotional tactics.

In this way, the methodological framework developed in this study not only optimizes short-term discount allocation but also provides insights to inform long-term brand strategy. The results indicate the importance of combining predictive analytics with strategic judgment to balance commercial performance and brand health. As customer expectations and competitive dynamics continue to evolve, such integrated approaches offer a direction for managerial practice.

# Appendices

## Appendix A Programming Code

Please find the full programming code of the research at [github.com/yoojinshim/thesis\\_code](https://github.com/yoojinshim/thesis_code)

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